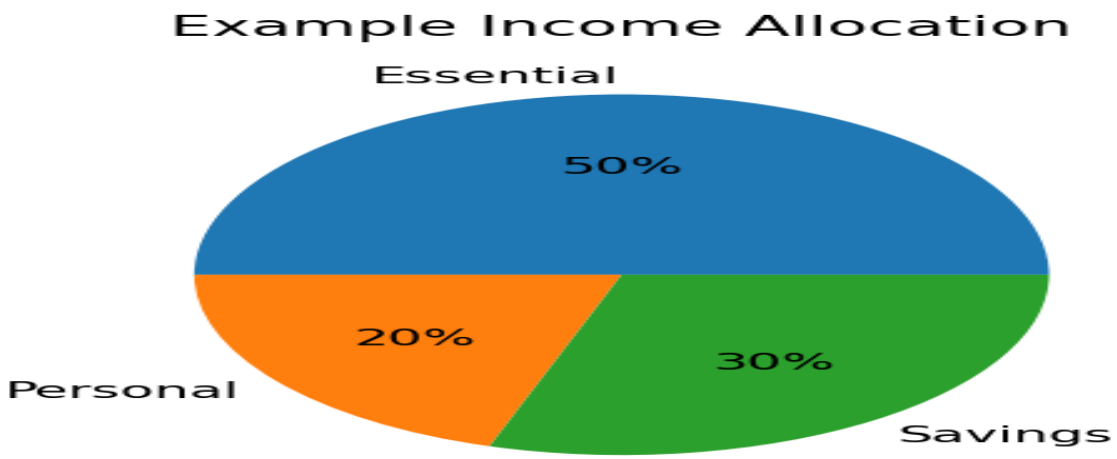
# 4.2 Smart Budget Planning

## Income-to-Expense Allocation

Income-to-expense allocation is the process of dividing available income into different financial categories. The Smart Budget Assistant can allocate money for essential expenses, personal spending, and savings. The percentages can be customized according to the user's situation. This gives users a clear view of where their income is going.

### Objectives

- Divide income into meaningful categories.
- Prioritize essential expenses.
- Keep a planned amount for savings.
- Control discretionary spending.
- Show the allocation clearly to the user.




### Working Procedure

1. Enter total monthly income.
2. Choose allocation percentages.
3. Calculate the amount for each category.
4. Check that the percentages total 100%.
5. Display the allocation and remaining amount.

### Example Data Table

| Category | Suggested Allocation | Example on ■30,000 |
|---|---|---|
| Essential Expenses | 50% | ■15,000 |
| Personal Spending | 20% | ■6,000 |

| Category | Suggested Allocation | Example on ■30,000 |
|---|---|---|
| Savings | 30% | ■9,000 |
| Total | 100% | ■30,000 |

## Python Implementation

```
        income = float(input("Enter monthly income: "))

        essential_percent = 50
        personal_percent = 20
        savings_percent = 30

        essential = income * essential_percent / 100
        personal = income * personal_percent / 100
        savings = income * savings_percent / 100

        print("\n--- Income Allocation ---")
        print("Essential Expenses:", essential)
        print("Personal Spending:", personal)
        print("Savings:", savings)
        print("Total Allocated:", essential + personal + savings)
```

## Benefits

Allocation provides a visual understanding of how income is distributed. Users can modify the percentages according to their own needs while maintaining awareness of essential costs and savings.

Note: All amounts are illustrative examples for academic demonstration. A real application can collect data through forms, a database, or transaction records.